

Renxiang Chu

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EDUCATION

Peking University

- Bachelor of Medicine in Basic Medical Sciences
Dual Degree in Economics

Beijing, China

Expected Jul. 2027

Core Courses:

Medical Molecular and Cell Genetics, Organic Chemistry, Nervous System, Introduction to Computing (B), Principles of Economics, Probability and Statistics (B), Macroeconomics and Health

RESEARCH EXPERIENCE

PKU Basic Medical Sciences Innovative Talent Program

Position: Research Trainee

Sept. 2023 - Aug. 2024 | Beijing

- Joined a bioinformatics research group to study the computational characterization of biomolecular condensate composition and dynamic regulation.
- Data Processing and Quantitative Analysis: Developed customized data-processing scripts in R and Python to quantitatively characterize the compositional complexity and phase-separation dynamics of biomolecular condensates.
- Mechanistic Validation: Analyzed the spatiotemporal regulation of liquid-liquid phase separation through literature synthesis and experimental validation, linking computational findings with biochemical assays to investigate the structural stability and functional roles of macromolecular assemblies.

Peking University Institute for Advanced Clinical Medicine

Position: Research Assistant

Supervisor: Meng Lu

Sept. 2024 - June 2027 | Beijing

- Joined the Artificial Intelligence and Dynamic Structure Laboratory to study applications of computer vision in biomedicine. Worked on AI applications in ophthalmology; helped initiate collaborations with Peking University First Hospital and Peking University People's Hospital to build a deep learning model for predicting biological age from fundus photographs.
- Foundation Model Engineering: Leveraged VisionFM (ViT + masked autoencoder, pre-trained on 3.4 million ophthalmic images) and fine-tuned a retinal aging clock on 112,019 fundus images, achieving state-of-the-art performance (MAE = 2.49 years).
- Latent Space Disentanglement: Designed a teacher-student residual learning framework to explicitly separate physiological aging features from pathological noise in latent space, enabling accurate biological age estimation even in patients with ocular comorbidities.
- The work was posted on medRxiv (co-first author), DOI: <https://doi.org/10.64898/2026.02.13.26346265>, with submission to a high-impact journal planned.
- Conducted research and design on an AI Co-scientist based on multi-agent systems.
- Designed and implemented SciEngram, a traceable cognitive memory architecture and evaluation benchmark for long-term scientific memory and cross-paper reasoning, enabling agents to accumulate paper evidence, organize cross-paper relationships, update current beliefs, and generate verifiable scientific conclusions.
- Designed a systematic benchmark covering factual retrieval, evidence attribution, cross-paper relationship judgment, conflict detection, scope constraints, unanswerable-query detection, and dynamic belief updating, with unified comparisons against long-context models, conventional RAG, and general-purpose memory systems.
- The manuscript was submitted to the AAAI 2027 Conference (co-first author).

Peking University School of Basic Medical Sciences

Position: Research Assistant

Supervisor: Guangxi Wang

Mar. 2026 - June 2027 | Beijing

- Joined the Institute of Systems Biomedicine for undergraduate thesis research, focusing on machine-learning- and deep-learning-based multi-omics modeling and contributing to an AI Co-Scientist workflow for automated cancer biomarker discovery.
- Conducted research on an AI-assisted medical education system (supervisors: Guangxi Wang and Xiaofan Wei). Designed and implemented HistoSeek, a teacher-governed AI-assisted histology learning platform, to address insufficient visual-localization training and personalized feedback in histology education. Conducted an exploratory study in an undergraduate histology course, comparing AI-assisted and conventional teaching in theoretical performance, specimen identification, and practical skills, and used learner questionnaires to assess usability, feedback value, and classroom acceptance.
- The manuscript was submitted to npj Digital Medicine (co-first author) and is currently under external review.

- Worked on a Northeastern University-MGH collaboration on AI world models and offline decision agents.
- Using MIMIC and a private heart-failure dataset, designed an action-conditioned world model for high-risk sequential decisions. It predicts future states and risks from history and candidate actions, providing a controllable environment for offline simulation, policy evaluation, and agent training.
- Contributed to model capability evaluation, including causal tracing and counterfactual reasoning.

PUBLICATIONS

Chu, R.#, Sun, A.#, Qu, J., & Lu, M. (2026). *Disentangling physiological heterogeneity in retinal aging using a deep learning-based biological age framework* (<https://doi.org/10.64898/2026.02.13.26346265>). *medRxiv* (Preprint)

Chu, R.#, Sun, C.#, Zhang, Y.#, Sun, Z.#, Chen, Y., Shang, Y., Jiang, Y., & Lu, M. (2026). *SciEngram: Evidence-grounded scientific memory for cumulative research reasoning*. Manuscript submitted to *AAAI 2027*

Chu, R.#, Chi, X.#, Li, Z., Zhang, X., Xu, J., Wang, G., & Wei, X. (2026). *HistoSeek links histology image recognition to teacher-authored formative guidance*. Manuscript under review at *npj Digital Medicine*

Equal contribution

PRACTICAL EXPERIENCE

Team Leader, Peking University School of Basic Medical Sciences Social Practice Group

- Organized the Peking University School of Basic Medical Sciences social practice delegation to Suzhou under the theme "Exploring Development Achievements, Fostering the Spirit of Innovation," and served as team leader.
- Visited Soochow University and Kunshan Biomedical Industrial Park and met with Synermore Biologic Co., Ltd.

SKILLS

- **Skills:** MS Office (Proficient); Python/R (Working Knowledge)
- **Tools & Platforms:** Linux, AWS, Conda, Jupyter, NumPy, Pandas, LaTeX, Git, Markdown
- **Languages:** English (CET-6: 607; TOEFL: 106); Mandarin Chinese (Native)